

What the guard bought

A bit-for-bit counterfactual of assertion-guarded refitting
in the block-1 positive control

Technical note

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Abstract

The block-1 positive control study (`code/analysis/time_block_forecast/block1_positive_control/`) fits 40 chains — settings $\{4, 5\} \times$ methods $\{\text{i.i.d.}, \text{AR}(2)\} \times$ datasets 1–10 — under an assertion guard: a fit that fails truth recovery (B) or predictive-spread (C) is refitted with a new seed, at most three times. The guard replaced the attempt-0 chain in 13 of the 40 cells. This note quantifies exactly what that intervention did to the forecast scores, by refitting every replaced attempt-0 chain bit-for-bit (same seed, same call) and scoring it — i.e. by reconstructing the study an *unguarded* pipeline would have delivered. On the affected cells the guard improved the Brier score by 16% and the CRPS by 12% on average; the rescues are largest exactly where the reflected-ridge law $\max_h \text{sd}(\hat{y}_h) \approx 1.5 |\hat{\lambda}|$ predicts (up to $12\times$ on Brier for $|\hat{\lambda}| \geq 5$), and negligible for mild reflections. One cell moved the other way: a $\hat{\lambda} = -15$ reflected chain *outscored* its healthy replacement. We argue this exception is not a defect but the point: the guard selects fits that are consistent with the model and the known truth — a criterion computable *before* seeing validation outcomes — and therefore it is not, and must not be, a score optimizer.

1 Setting

The positive control asks whether fitting AR(2)-generated data with the AR(2) model beats an i.i.d.-in-time skew-t baseline in forecast scores. Design: data settings 4 (strong, $\phi = (0.80, -0.35)$, spectral radius 0.592) and 5 (near-unit-root, $\phi = (0.15, 0.80)$, radius 0.973); datasets 1–10; block 1 only (train $t = 1..50$, forecast $t = 51..65$); both methods. Truth: $\lambda = 3$, $\beta_0 = 10$.

Block 1’s short prefix is where the reflected ridge (`tex/lambda_phiz_ridge`) bites hardest, so every fit runs the self-consistency assertions of `fit_diag_utils.R` and is kept only if it passes

B truth recovery: $\text{sign}(\hat{\lambda}) = +$, $|\hat{\lambda}/3 - 1| < 0.6$, $\hat{\beta}_0$ near 10;

C predictive spread: $\max_h \text{sd}(\hat{y}_h) \leq 3 \times \text{marginal SD}$,

otherwise it is refitted with seed $+7919 \cdot \text{attempt}$, at most three reseeds, *symmetrically for both methods* (the ridge is a property of the shared skew-t family: the i.i.d. method reflected five times in this study). A cell still failing after four attempts is kept but flagged, and the primary analysis excludes it. The rule was pre-registered before any scores were computed.

The guard changed the delivered chain in 13 cells: 10 rescued (reseed found a passing chain) and 3 flagged (setting 5: dataset 2 under both methods, and dataset 8 under AR(2)).

2 The counterfactual

`run_block1_guarded.R` saves only the kept fit, so the unguarded study is not on disk — but it is fully reproducible: the attempt-0 seed is `get_tbf_seed(setting, method, dataset)`, and rerunning `mcmc()` with it reproduces the rejected chain bit-for-bit. `guard_effect.R` does exactly this for the 13 affected cells, pushes each chain through `forecast_block()`, and scores both versions under the study protocol: CRPS and the Brier score at the q_{95} threshold of the validation block, averaged over leads 1–5 (the pre-registered primary window). Define

unguarded study = attempt-0 chain in every cell, guarded study = the kept chains.

The 27 unaffected cells are identical in both, so the whole effect of the guard lives in Table 1.

3 Results

Table 1: All 13 affected cells. m1 = i.i.d., m2 = AR(2); arrows read attempt-0 $\rightarrow$ kept. “att” = attempts used (4 = flag kept). Scores are means over leads 1–5.

cell		att	$\hat{\lambda}_0$	$\hat{\lambda}_{\text{kept}}$	CRPS		Brier q_{95}	
					att-0	kept	att-0	kept
s4 d5 m2	rescue	2	−7.28	+3.93	2.64	1.57	0.0205	0.0017
s4 d5 m1	rescue	2	−7.23	+3.52	3.19	1.90	0.0265	0.0033
s4 d10 m1	rescue	2	−4.97	+2.99	2.87	2.21	0.0558	0.0226
s5 d7 m1	rescue	2	−3.99	+2.47	1.61	1.29	0.0832	0.0463
s4 d4 m2	rescue	2	−4.01	+2.19	1.46	1.12	0.0010	0.0022
s4 d6 m2	rescue	3	−8.90	+2.86	5.59	5.48	0.1378	0.1349
s5 d5 m1	rescue	2	−4.28	+2.60	2.80	2.45	0.0164	0.0166
s5 d6 m2	rescue	2	+5.40	+3.95	2.24	2.30	0.0794	0.0799
s5 d8 m1	rescue	4	−2.62	+2.29	1.97	2.37	0.0000	0.0007
s5 d3 m2	rescue	4	−15.18	+3.34	1.94	2.15	0.1088	0.1323
s5 d2 m1	flag	4	+0.23	+0.24	1.17	1.17	0.0194	0.0194
s5 d2 m2	flag	4	+0.55	+0.56	1.16	1.16	0.0190	0.0190
s5 d8 m2	flag	4	−2.47	−2.54	1.23	1.25	0.0000	0.0000

Table 2: Group means over the affected cells (leads 1–5).

group	cells	CRPS		Brier q_{95}	
		att-0	kept	att-0	kept
setting 4, i.i.d.	2	3.03	2.05	0.0412	0.0129
setting 4, AR(2)	3	3.23	2.73	0.0531	0.0463
setting 5, i.i.d.	4	1.89	1.82	0.0298	0.0207
setting 5, AR(2)	4	1.64	1.71	0.0518	0.0578
all affected	13	2.30	2.03	0.0437	0.0368

Three regularities.

1. The rescue size follows the $|\lambda|$ law. The ridge note established $\max_h \text{sd}(\hat{y}_h) \approx 1.49 |\hat{\lambda}|$ across every fit, healthy or broken. Since an overdispersed, mis-centred predictive distribution is punished by both proper scores, the damage of a reflected chain — and hence the value of replacing it — should grow with $|\hat{\lambda}_0|$. It does. The two $|\hat{\lambda}_0| \approx 7$ reflections (s4 d5) give the largest rescues: Brier $0.0205 \rightarrow 0.0017$ and $0.0265 \rightarrow 0.0033$, i.e. 8–12 $\times$; CRPS drops by 40%. The mild $|\hat{\lambda}_0| \approx 2.5$ reflections (s5 d8) score essentially the same as healthy chains — a weak reflection widens the forecast too little to matter. Both methods benefit: the i.i.d. method’s two setting-4 reflections were among the worst, and their repair drives that group’s Brier from 0.0412 to 0.0129.

2. Overall, honesty was cheap and profitable. Across the 13 cells the guard improved Brier by 16% and CRPS by 12%; folded into the full 40-cell study the guarded version is strictly better calibrated *and* slightly better scoring. The positive-control conclusions (AR(2) beats i.i.d. on CRPS at near-unit-root persistence, –31% at leads 1–5, Wilcoxon $p = 0.040$) were additionally shown to be insensitive to the flag-exclusions: the sensitivity run that keeps all cells gives $p = 0.010$ –0.021.

3. One cell went the other way, and it is the most instructive row in the table. In s5 d3 the AR(2) attempt-0 chain sat at $\hat{\lambda} = -15.18$ — the most extreme reflection in the study — yet scored *better* than its healthy replacement (CRPS 1.94 vs 2.15, Brier 0.109 vs 0.132). Dataset 3 is the hardest in setting 5 (both methods score Brier > 0.1 there even when healthy); on a hard validation window, an overdispersed forecast hedges, and a confident, well-calibrated forecast pays for its confidence. A finite window of 15 days can and occasionally will reward the wrong model.

4 Why the exception is the point

Suppose the guard had been built the obvious wrong way: refit until the *score* improves. Then s5 d3 would have kept the $\hat{\lambda} = -15$ chain — a fit whose skew loading has the wrong sign by a factor of five, whose latent path is upside down, and whose predictive spread is inflated 5 $\times$ — because it happened to win on one 15-day window. That is selection on outcomes: it biases every downstream comparison and cannot be defended.

The actual guard never looks at validation data. Assertions B and C are computable from the fit and the training data alone (B uses the simulation truth, which a simulation study owns by construction; C uses the climatology bound that any stationary forecast must respect). The guard therefore implements *selection on model consistency*: it asks “is this chain a sample from the posterior the model claims?”, not “did this chain get lucky?”. The price is occasionally visible — about +0.02 Brier on one cell here — and it is the premium for an unbiased study. The correct reading of Table 1 is not “the guard won 9 rows and lost 1” but “the guard delivered 37 of 40 cells with certified-consistent fits, and told us exactly which 3 it could not certify.”

The flagged cells carry information too. Dataset 2 of setting 5 failed in the same way under *both* methods and all four seeds: $\hat{\lambda} \approx 0.25$ (i.i.d.) and ≈ 0.56 (AR(2)), stable to three decimals across reseeds. This is not chain capture — it is the *data*: under a near-unit-root z process ($h^* \approx 109$ days), a single 50-day training window can realize an essentially flat z path, in which case λ is genuinely under-identified by that window and no reseed can fix it.

The A' statistic separates the two situations cleanly in its first field test: passing fits have $sd_t(\hat{z})/(\hat{\sigma}\sqrt{1-2/\pi}) \approx 1.04\text{--}1.17$, flagged fits $0.57\text{--}0.77$. Flagging, rather than silently averaging such cells into the comparison, is the guard doing for the study what assertions A/C did for the `z.init` bug: converting a quiet distortion into a visible, attributable one.

5 Summary

- An unguarded block-1 study would have shipped 10 reflected or inflated chains among 40; the guard replaced them at a cost of ~ 1 extra fit per affected cell.
- Effect on scores, affected cells: Brier -16% , CRPS -12% ; rescue magnitude $\propto |\hat{\lambda}_0|$, up to $12\times$ on Brier, negligible below $|\hat{\lambda}_0| \approx 3$.
- Both methods needed it: the ridge is a family property, not an AR(2) property.
- One cell paid a small score premium for consistency; that premium is the operational difference between guarding on *assertions* and cherry-picking on *outcomes*.
- The three flagged cells are a data-level identifiability statement about short windows under near-unit-root persistence, now visible instead of silent.

A guard that can never lose points against an oracle is not a guard; it is a leaderboard. The value of assertion-guarded refitting is that its selections are defensible before the validation data are opened.

A Literature-search prompt

The following self-contained prompt packs the reflected-ridge phenomenon of the companion note `tex/lambda.phiz.ridge` and the empirical evidence of this note into a form that can be pasted into a textbook database, a search engine, or another assistant, to locate the established terminology and standard treatments of the problem. It is deliberately written in plain ASCII with no project-specific jargon.

I observed the following phenomenon in a Bayesian hierarchical skew-t spatio-temporal model and want to connect it to the established literature. Please give me (a) the canonical / professional names of each part of the phenomenon, (b) textbook or review-paper treatments, (c) the same problem in other model classes, and (d) the named standard remedies.

MODEL. Observations

$y_{\{st\}} = \text{beta0} + x'\text{beta} + \text{lambda} * z_t + \text{eps}_{\{st\}}/\text{sqrt}(\text{tau_t})$,
 where $z_t \geq 0$ is a latent half-normal factor, $z_t \sim \text{HN}(0, \text{tau_t}^{-1/2})$, given AR(2) temporal dynamics through a Gaussian copula; lambda is a scalar skewness loading with a diffuse sign-symmetric prior $N(0, 20)$. Simulation truth: $\text{lambda} = +3$, $\text{beta0} = 10$. Fitting is by MCMC (Gibbs/Metropolis), 20k iterations.

CORE GEOMETRY. The likelihood depends on $(\text{beta0}, \text{lambda}, \{z_t\})$ only through the sum $m_t = \text{beta0} + \text{lambda} * z_t$. For any fitted trajectory $\{m_t\}$, every pair $(\text{beta0}*, \text{lambda}*)$ with the

reassignment $z_{t*} = (m_t - \beta_0)/\lambda$ reproduces the likelihood EXACTLY, provided $z_{t*} \geq 0$. The likelihood is flat on this two-parameter family; identification comes only from the prior structure on z (its half-normal marginal and its temporal smoothness). The family has two branches:

- positive branch ($\lambda > 0$): z path keeps its shape; posterior mass near the truth;
- reflected branch ($\lambda < 0$): z path is flipped upside down, and β_0 must rise above $\max_t m_t$ to keep $z \geq 0$.

OBSERVED CONSEQUENCES (evidence from 25 + 40 controlled refits on simulated data with known truth):

1. A subset of chains (7/25, later 13/40) settle on the reflected branch: λ flips sign to $-4..-15$, β_0 rises to $19..35$ against a data mean of ~ 14.5 , so $\beta_0 + \lambda * \text{mean}(z)$ still reproduces the data mean exactly.
2. On that branch the temporal variation of z shrinks like $1/|\lambda|$, so the fitted latent path is far too smooth for its own half-normal marginal; the AR coefficients drift to a near-unit root, which makes the over-smooth path cheap under the prior (the dynamics "absorb" the violation).
3. A first-moment check on z (mean of z vs its model-implied value) PASSES on every reflected fit -- the ridge preserves the first moment by construction. A second-temporal-moment check, $\text{sd}_t(z) / (\sigma * \sqrt{1 - 2/\pi})$, separates cleanly: healthy fits $1.04..1.17$, pathological fits $0.57..0.77$.
4. Out-of-sample forecasts amplify the hidden inconsistency: $\max$ predictive SD = $1.49 * |\hat{\lambda}|$ ($R^2 = 0.987$ through the origin, healthy and pathological fits on the same line), because forecasting redraws z from the model law at full temporal variance and multiplies by the inflated $|\lambda|$.
5. The posterior is bimodal and the chain cannot switch modes: leaving the reflected branch requires moving λ across zero AND dropping β_0 by ~ 20 AND re-inverting the whole z path simultaneously; every one-coordinate move is individually catastrophic for the likelihood.
6. Restarting the SAME data with a new RNG seed rescues most cases (10/13), showing the capture is chain-level, i.e. decided by initialization during burn-in. In a few short-window datasets the failure REPRODUCES across four seeds with λ stable to three decimals: there the training window realized a nearly flat latent path and λ is genuinely under-identified by the data, not by the sampler.
7. Guarding the study with pre-registered consistency assertions (computable from fit + training data only, never validation scores) and refitting failed cells with deterministic new seeds repaired the study; in one cell the pathological fit accidentally SCORED BETTER out of sample than its healthy replacement, which is why the guard must select on model consistency, not on outcomes.

WHAT I SUSPECT THIS RELATES TO (please confirm, correct, and name precisely):

- sign/reflection non-identifiability of factor loadings in factor analysis (λ, z) $\rightarrow$ $(-\lambda, -z)$, here modified by the positivity constraint $z \geq 0$ into $(\lambda, \beta_0, z) \rightarrow (-\lambda, \beta_0 + \text{shift}, \text{flipped } z)$;
- likelihood ridges / flat directions; weak identification; aliasing; additive confounding of intercept and latent level (as in spatial confounding?);
- skew-normal / skew-t identifiability problems (Azzalini; singular information at $\lambda = 0$, centered vs direct parameterization);
- label switching in mixtures (an analogous discrete symmetry, but for component labels rather than a sign);
- multimodal posteriors and mode capture / poor mixing between symmetric modes; tempering and mode-jumping proposals;
- anchoring / sign constraints / positive-loading identification conventions in Bayesian factor models;
- non-centered vs centered parameterizations of hierarchical models.

QUESTIONS.

1. What is the accepted name for the joint sign-reflection + compensating-location-shift invariance above, in the factor analysis / skew-elliptical literature? Which textbook treats it most directly?
2. Is there an established name/theory for diagnostics that lie ALONG a ridge (blind to it) versus TRANSVERSE to it (detect it) -- i.e. for choosing model-checking statistics relative to a non-identified direction?
3. What are the named remedies and their trade-offs: sign constraints/anchoring, asymmetric priors, reparameterization, mode-relabeling in post-processing, mode-jumping MCMC?
4. Is there literature on the validity of restart/reseed-until-consistent policies (selection on model consistency rather than on outcomes), and on the selection bias induced if one instead restarts until the score improves?
5. Is the empirical linear law "predictive SD proportional to |loading|" a known signature of loading-inflation pathologies in latent factor forecasts?